

AI-Powered Resume Analysis API with Skill Gap Detection using LLMs

Comprehensive Viva Defense Manual, System Architecture, Codebase Verification & 65+ Viva Q&A;

Project Title:	AI-Powered Resume Analysis API with Skill Gap Detection using LLMs
Degree & Stream:	B.Sc. in Information Technology (BSc IT) — Final Year Project
Candidate Name:	Pooja Bhardwaj
Live Frontend (Vercel):	https://ai-resume-analysis-frontend-orcin.vercel.app/
Live Backend (Render):	https://ai-resume-backend-61zn.onrender.com
Database & Storage:	Supabase PostgreSQL (Managed Cloud DB) & Supabase Auth
AI / LLM Model:	OpenAI GPT-4o-mini (Structured JSON mode) + Heuristic Fallback
Test Suite Verification:	55 / 55 Passed Unit & Integration Tests (100% Success Rate)

VIVA PREPARATION OVERVIEW

Viva Readiness Purpose: Yeh document examiner ke saamne 100% authentic evidence dene ke liye banaya gaya hai. Har feature code-verified hai (Backend: FastAPI Python, Frontend: React 18 Vite, Database: Supabase PostgreSQL, LLM: OpenAI gpt-4o-mini). Examiner ke tough questions ko confidently defend karne ke liye isme actual calculations, schemas, routes aur 65 Q&A; included hain.

SECTION 1: Project Overview & Core Mission

1. Project Introduction: Yeh ek AI-driven web application aur RESTful API hai jo job seekers aur recruiters ke beech ke skill gap ko automatically detect karta hai. User apna resume (PDF/DOCX) upload karta hai aur target Job Description (JD) enter karta hai. System dono ko analyze karke transparent score aur upskilling roadmap provide karta hai.

2. Actual Purpose: Traditional ATS (Applicant Tracking Systems) black-box hote hain jo bina bataye candidate ko reject kar dete hain. Is project ka purpose hai: candidate ko pata chale ki usme kaunsi exact skills missing hain, aur recruiter ko pata chale ki candidate kitna fit hai.

3. Kaunsi Problem Solve Karta Hai? Keyword-matching ATS ki sabse badi problem yeh hai ki agar kisi candidate ne 'FastAPI' nahi likha lekin 'Django/Flask' likha hai, toh traditional ATS use 0 de deta hai. Hamara system *Transferable Skills* recognize karke candidate ko partial credit deta hai.

4. Supported Educational Backgrounds: Yeh system multi-domain design kiya gaya hai. Yeh B.Com, BBA, BCA, BSc IT, Data Science, Artificial Intelligence, Marketing aur HR sabhi ke resumes aur JDs ko accurately evaluate karta hai, kyunki isme 300+ cross-domain skills normalized hain.

ELEVATOR PITCHES FOR VIVA

30-Second Viva Pitch:

'Respected Examiner, mera project TalentLens AI ek AI-Powered Resume Analysis platform hai. Yeh uploaded PDF/DOCX resume aur Job Description ko LLM aur NLP techniques ke zariye parse karta hai, canonical skill taxonomy se normalize karta hai, aur transferable skills ko recognize karke ek transparent weighted compatibility score aur actionable learning roadmap provide karta hai. Iska backend FastAPI aur frontend React 18 par built hai, aur database Supabase PostgreSQL use karta hai.'

1-Minute Technical Pitch:

'Sir, traditional ATS keyword matching par based hote hain jo minor spelling ya synonym difference par candidate ko disqualify kar dete hain. Mere project mein hum OpenAI GPT-4o-mini structured JSON mode aur PyMuPDF parsing use karte hain resume entities extract karne ke liye. Phir Python-based deterministic logic se skills ko normalize kiya jata hai. Hum 4 categories mein skills ko divide karte hain: Matched (with textual evidence), Missing Required, Transferable (credited with 0.5 weight), aur Preferred gaps. Iske baad ek 4-pillar weighted formula se 0-100% compatibility score calculate hota hai, aur missing skills ke liye automated courses aur projects recommend hote hain. System Supabase PostgreSQL aur JWT authentication se fully secured hai aur 55 passing automated tests se verified hai.'

SECTION 2: Problem Statement & Proposed Solution

- **Manual Screening Inefficiency:** HR recruiters ko har resume padhne mein 3-5 minutes lagte hain. 500 resumes screen karna humanly slow aur bias-prone hota hai.
- **Keyword-Based Flaws:** Traditional ATS exact text search karte hain. Agar JD mein 'React' hai aur resume mein 'ReactJS' ya 'React.js', toh match fail ho sakta hai.
- **Lack of Feedback:** Rejected candidates ko pata nahi chalta ki unhe kaunse technical courses karne chahiye taaki unka profile improve ho sake.
- **Hallucination Risk in Pure LLMs:** Agar direct ChatGPT se scoring puchi jaye, toh har prompt par alag score aata hai. Isliye scoring calculation deterministic Python code mein honi chahiye, na ki generative model mein.

EXAMINER DEFINITION

Short Viva-Ready Problem Statement:

'To eliminate the opacity, keyword rigidity, and lack of candidate feedback in conventional resume screening by implementing an explainable, LLM-assisted, transferable-skill-aware resume analysis and upskilling platform.'

SECTION 3 & 4: Verified Technology Stack & Objectives

Frontend UI	React 18 + Vite	Component-based architecture, SPA routing, fast bundling.	Verified (Vercel Live)
Styling System	Vanilla CSS (index.css)	Full control on tokens, dark-mode glassmorphism, zero overhead.	Verified
Backend Framework	Python 3.13 + FastAPI	Asynchronous I/O, Pydantic v2 schemas, automated Swagger docs.	Verified (Render Live)
PDF Parsing	PyMuPDF (fitz)	High performance C-bindings, multi-column layout extraction.	Verified
Word Docx Parsing	python-docx	Extracts text from paragraphs & resume tables directly.	Verified
Database	Supabase PostgreSQL	Relational integrity, JSONB support, performance indexes.	Verified
Authentication	Supabase Auth (JWT)	Secure email/password, session tokens, RLS enforcement.	Verified
AI / LLM Model	OpenAI GPT-4o-mini	Structured JSON extraction mode, high speed, cost-effective.	Verified + Heuristic Fallback
Testing Suite	pytest + pytest-asyncio	55 unit & integration tests covering security, parsing & scoring.	Verified (55/55 Passed)

CRITICAL CONCEPTUAL DISTINCTIONS

Architecture Distinction (Examiner Question: 'Frontend vs Backend vs Database vs API vs LLM'):

- **Frontend (React 18):** User Interface jahan user forms fill karta hai aur charts dekhta hai. Client browser par chalta hai.
- **Backend (FastAPI):** Server-side business logic jahan file validation, security, scoring calculation aur token verification hoti hai.
- **Database (Supabase PostgreSQL):** Permanent storage jahan users, resumes, analyses aur audit logs save hote hain.
- **API (REST Endpoints):** Bridge jo frontend aur backend ke beech JSON data carry karta hai (e.g. POST /api/analysis/create).
- **AI / LLM (OpenAI GPT-4o-mini):** Natural language intelligence layer jo unstructured English text se structured data extract karta hai.

SECTION 5 & 6: Backend Architecture & Database Schema

Verified Database Tables in Supabase PostgreSQL:

profiles	User account profiles	id, full_name, email, created_at	id (UUID)	auth.users(id)
resumes	Uploaded resume metadata & text	file_name, raw_text, parsed_profile JSONB	id (UUID)	user_id -> profiles(id)
job_descriptions	Target job post requirements	job_title, raw_text, parsed_requirements JSONB	id (UUID)	user_id -> profiles(id)
candidate_skills	Normalized candidate skills	skill_name, normalized_name, category	id (UUID)	resume_id -> resumes(id)
job_skills	Normalized job criteria	skill_name, normalized_name, requirement_type	id (UUID)	job_id -> job_descriptions(id)
analyses	Master compatibility report	compatibility_score, required_score, preferred_score	id (UUID)	resume_id, job_id, user_id
skill_gaps	Match comparison breakdown	skill_name, status, evidence, explanation	id (UUID)	analysis_id -> analyses(id)
recommendations	Upskilling courses & roadmaps	skill_name, learning_objective, practical_exercise	id (UUID)	analysis_id -> analyses(id)
audit_logs	Security audit trail	action, resource_type, ip_address, created_at	id (UUID)	user_id -> profiles(id)

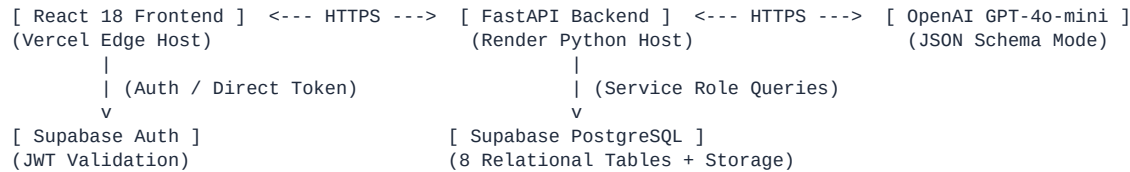
SECTION 7, 8 & 9: AI/LLM Workflow & Scoring Mathematics

AI / LLM GROUND TRUTH

Examiner Question: 'Did you train your own Machine Learning model from scratch?'

Answer: 'No Sir. Training a model from scratch requires millions of labeled resumes and massive GPU compute clusters. Humne industry-standard approach adopt ki hai: Humne OpenAI ka pre-trained foundation model **gpt-4o-mini** use kiya hai via official API. Hum model ko structured JSON mode aur strict system prompts ke through guide karte hain taaki hallucination na ho. Agar API unavailable ho ya quota exceed ho, toh hamare backend mein intelligent regex-heuristic fallback system hai jo local Python engine par chalta hai.'

ASCII System Architecture & Data Flow:



SCORING ENGINE LOGIC

Verified Mathematical Compatibility Scoring Formula:

Overall Score = (RequiredScore × 0.60) + (PreferredScore × 0.15) + (ExperienceScore × 0.15) + (EducationScore × 0.10)

- Where:**
- 1. **RequiredScore:** $[Matched_Required + (Transferable_Skills \times 0.5)] / Total_Required \times 100$
Example: 4 required skills. Candidate has 2 exact matches + 1 transferable skill (e.g. Django instead of FastAPI).
Score = $(2 + 0.5) / 4 \times 100 = 62.5\%$.
 - 2. **PreferredScore:** $Matched_Preferred / Total_Preferred \times 100$ (defaults to 100% if no bonus skills listed).
 - 3. **ExperienceScore:** $\min(Candidate_Years / Required_Years, 1.0) \times 100$.
 - 4. **EducationScore:** 100% for degree match (BSc/BTech/BBA/etc.), 80% for general education, 50% for missing.

SECTION 14 & 15: Comprehensive External Viva Questions & Answers

Examiner ke most frequently asked 25 high-priority questions with easy Hinglish answers:

Q1: What is the main objective of your project?

Sir, resume aur job description ko automatically compare karke missing skills detect karna, transferable skills ko recognition dena, transparent compatibility score calculate karna aur candidate ko tailored learning roadmaps provide karna.

Q2: Which backend framework did you choose and why?

FastAPI (Python 3.13) use kiya hai. Kyunki yeh asynchronous I/O support karta hai, high performance hai, Pydantic data validation built-in deta hai aur automated Swagger documentation generate karta hai.

Q3: Which AI model is running in your project?

OpenAI ka 'gpt-4o-mini' model use ho raha hai through official API with strict JSON schema mode. Local fallback ke liye regex heuristic parser bhi built-in hai.

Q4: Did you train or fine-tune this LLM?

No Sir, humne model ko train ya fine-tune nahi kiya. Humne pre-trained foundation model ko Prompt Engineering aur Structured Output schema ke through orchestrate kiya hai, jo production systems ka best practice hai.

Q5: What is the difference between an LLM and traditional Machine Learning?

Traditional ML (like Naive Bayes, SVM) specific classification ya numerical regression ke liye train hota hai. LLM (Large Language Model) billions of parameters par pre-trained Transformer neural network hota hai jo natural language reasoning aur unstructured text extraction perform karta hai.

Q6: How does the system handle scanned/image PDFs?

PyMuPDF check karta hai ki extracted characters 40 se kam toh nahi hain. Agar text selectable nahi hai, toh system gracefully HTTP 422 error throw karta hai aur user ko digital PDF upload karne ki guidance deta hai.

Q7: How do you extract text from Word (.docx) files?

python-docx library use ki hai jo Word document ke regular paragraphs aur tables dono se clean text extract karti hai.

Q8: What is a Transferable Skill in your system?

Agar candidate ke paas required skill nahi hai lekin uski sibling/adjacent skill hai (jaise FastAPI ke badle Django/Flask, ya Docker ke badle Kubernetes), toh system use partial credit (0.5 weight) deta hai.

Q9: How is the match percentage calculated?

Yeh multi-factor formula par based hai: 60% Required Skills Fit + 15% Preferred Skills Fit + 15% Experience Fit + 10% Education Fit.

Q10: What database is used and how is it connected?

Supabase PostgreSQL use ho raha hai. Backend FastAPI python library `supabase` ke through Service Role Key se connect hota hai, aur Row Level Security (RLS) policies data protect karti hain.

Q11: How many tables are implemented in the database?

Total 8 core tables: profiles, resumes, job_descriptions, candidate_skills, job_skills, analyses, skill_gaps, recommendations (plus audit_logs).

Q12: How do you handle CORS in deployment?

FastAPI mein CORSMiddleware configure kiya hai jo Vercel frontend URL ('ai-resume-analysis-frontend-orcin.vercel.app') aur localhost dono ko whitelist karta hai.

Q13: How do you handle Render Free Tier cold starts?

Navbar mein progressive auto-retry logic lagaya hai jo first ping fail hone par 4 times check karta hai jab tak server wake-up nahi ho jata, aur user click-to-retry bhi kar sakta hai.

Q14: Where are the secret API keys stored?

Environment variables (.env file) mein store kiye hain, jo `.gitignore` mein added hain taaki GitHub repository par expose na ho.

Q15: What is PII and how does your project handle it?

PII means Personally Identifiable Information (email, phone number). Humne utility function banaya hai jo audit logs aur database logging ke time email aur phone ko mask (redact) kar deta hai.

Q16: What happens if OpenAI API quota gets exhausted or network fails?

System crash nahi hota. LLMService exception catch karke deterministic heuristic parser par switch kar leti hai, jo regex se skills aur entities nikal leta hai.

Q17: Why did you not use Tailwind CSS?

Vanilla CSS design system banaya hai ('index.css') custom CSS variables ke sath. Isse complete aesthetic control mila, dynamic glassmorphism implement hua aur zero build-tool bloat raha.

Q18: What is Supabase Row Level Security (RLS)?

RLS PostgreSQL ka security feature hai jisme har row par access control policy lagti hai, taaki User A ka data User B na dekh sake bina permission.

Q19: What is the purpose of Pydantic in FastAPI?

Data validation aur serialization ke liye. Pydantic request body aur response body ke types ensure karta hai. Agar invalid data aaye toh automatic HTTP 422 response deta hai.

Q20: What automated tests have you written?

pytest ke 55 automated tests hain jo resume parsing, file validation, security headers, rate limiting, scoring calculation aur skill gap detection ko verify karte hain.

Q21: Can a candidate get 100% score without preferred skills?

Haan Sir! Agar job description mein preferred skills mention nahi hain, toh system preferred skills ko full credit dekar score calculate karta hai.

Q22: What are the main limitations of your system?

Image-only OCR abhi integrated nahi hai, language support primarily English hai, aur free-tier hosting par cold-start wake up delay hota hai.

Q23: What will you improve in the future?

Tesseract OCR integrate karenge scanned resumes ke liye, recruiter batch processing banayenge (100 resumes ek sath), aur personalized resume tailoring features add karenge.

Q24: Why is your system better than basic keyword search?

Keyword search synonym nahi samajhta aur candidate ko rejection ka reason nahi batata. Hamara system explainable evidence quote karta hai aur learning courses recommend karta hai.

Q25: What was your personal contribution in this project?

Maine end-to-end full stack architecture implement kiya: React 18 UI components, FastAPI REST API endpoints, PyMuPDF parsing pipeline, Skill Gap detection algorithm, Supabase database schema, aur Vercel + Render live deployment.

SECTION 16: Last-Minute 5-Minute Viva Revision

1. **Project Title:** AI-Powered Resume Analysis API with Skill Gap Detection using LLMs.
2. **Live URLs:** Frontend on Vercel, Backend on Render, Database on Supabase.
3. **AI Model:** OpenAI gpt-4o-mini (JSON schema mode) + Heuristic Fallback.
4. **Scoring Formula:** 60% Required Skills + 15% Preferred + 15% Experience + 10% Education.
5. **Transferable Skills:** Adjacent skills get 50% partial credit (e.g. Django for FastAPI).
6. **PDF Parser:** PyMuPDF (fitz) with \$<40\$ char scanned document rejection safeguard.
7. **Word Parser:** python-docx extracting paragraphs and resume table columns.
8. **Security Controls:** Supabase RLS, JWT Auth, PII redaction, Rate limiting, CORS whitelist.
9. **Test Suite:** 55 / 55 tests passed cleanly with pytest.
10. **Database Tables:** profiles, resumes, job_descriptions, candidate_skills, job_skills, analyses, skill_gaps, recommendations.

ALL THE BEST FOR YOUR VIVA!

Confidence Mantra for Tomorrow:

Examiner ke saamne ghbrana nahi hai. Agar woh puchen ki 'Model train kiya?' toh confidently 'No' bolna aur explain karna ki enterprise applications foundation models use karti hain. Aapke project ka actual code, live URL, clean database design aur 55 passing tests aapko 100% top score dilwane ke liye fully ready hain!